NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 27/03/2026
Time: 3 Hrs | GTM – 06 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	D	D	A	C	D	B	B	B	A
11	12	13	14	15	16	17	18	19	20
B	C	C	A	C	A	D	D	A	A
21	22	23	24	25					
25	96	6	91	9					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	D	D	D	C	A	B	C	B	D
36	37	38	39	40	41	42	43	44	45
C	A	A	D	D	B	D	A	C	D
46	47	48	49	50					
3	6	3	462	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	C	B	A	D	B	C	B	B	A
61	62	63	64	65	66	67	68	69	70
D	D	C	A	C	B	D	A	A	D
71	72	73	74	75					
21	1	9	1	1					

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HINTS
PHYSICS

1. Angular momentum--- ML^2T^{-1} Latent heat----L^2T^{-2} Torque---ML^2T^{-2} Capacitance----$ML^3T^{-1}Q^{-2}$

2. $\frac{4}{3}\pi R^3 = n\cdot\frac{4}{3}\pi r^3$

$n = \frac{R^3}{r^3}$ (i)

$\Rightarrow \quad W = T\left(n\cdot 4\pi r^2 - 4\pi R^2\right)$

$H = \frac{W}{J} = ms\Delta\theta = \frac{T}{J}\left(\frac{R^3}{r^3}\cdot 4\pi r^2 - 4\pi R^2\right)$

$\frac{H}{V} = \frac{3T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$

3. $S_2P - S_1P = \sqrt{5}d - 2d = \frac{\lambda}{2}$

$d = \frac{\lambda}{2\left(\sqrt{5}-2\right)}$

4. $T = 2\pi\sqrt{\frac{r^3}{GM}}$

$T_A = 2\pi\sqrt{\frac{\left[(6400+600)\times 10^3\right]^3}{GM}}$

$T_B = 2\pi\sqrt{\frac{\left[(6400+1600)\times 10^3\right]^3}{GM}}$

5. Direction of propagation of EM-wave = $\vec{E}\times\vec{B}$

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$$B = \frac{E}{C} = \frac{6}{3\times10^{8}} = 2\times10^{-8}T$$

6. Linear momentum conservation:

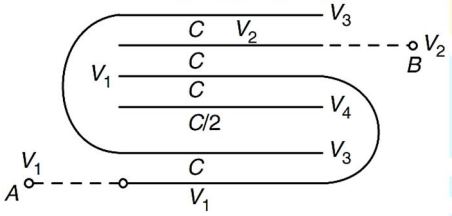
$$3u = 2v_2 + v_1$$

$$120 = 2v_2 + 60$$

$$v_2 = 30\,m/s$$

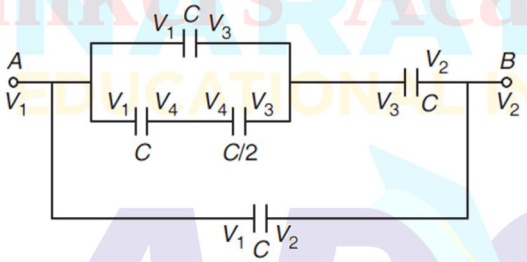
Fraction change in KE = $\frac{K_f - K_i}{K_i} = \frac{3600+1800-4800}{4800} = \frac{1}{8}$

7.



$$C = \frac{\varepsilon_0 A}{d} = 7\,\mu F$$

The equivalent circuit is as shown in figure.



$$C_{AB} = \frac{11}{7}C = \frac{11}{7}(7\mu F)$$

$$= 11\mu F$$

8. $B = \frac{\mu_0 i}{2r}$; magnetic field at centre of hydrogen atom i.e. at nucleus.

$$i = \frac{e}{T} = ef = \alpha z^2 / n^3$$

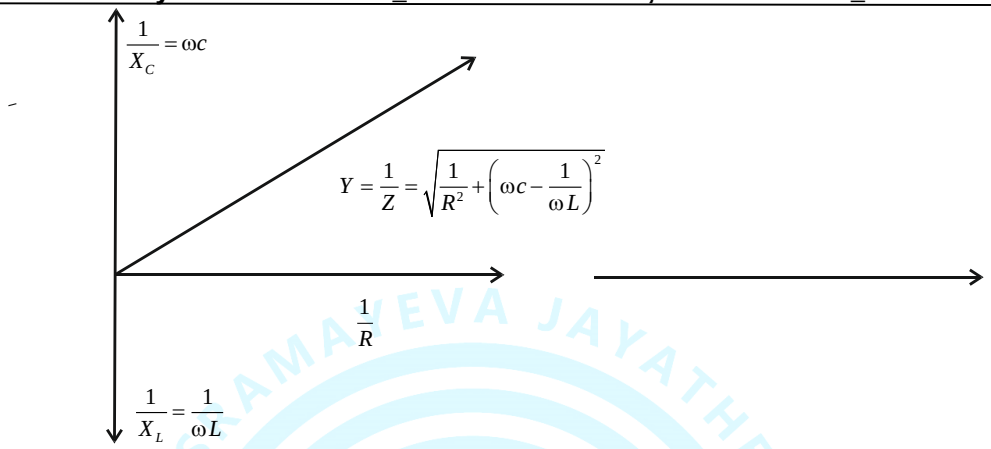
$$r \alpha n^2 / z$$

$$B \alpha i / r \alpha Z^3 / n^5$$

$$B \alpha 1 / n^5$$

9. Phasor of current taking potential as base line

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Admittance of circuit $= \sqrt{\frac{1}{R^2} + \left(\omega c - \frac{1}{\omega L}\right)^2}$

$$I_0 = \frac{V_0}{Z} = V_0 Y$$

$$= V_0 \sqrt{\frac{1}{R^2} + \left(\omega c - \frac{1}{\omega L}\right)^2}$$

10. $\square K = \int_2^4 Pdt = [t^3 - t^2 + t]_2^4 = 46J$

11. Let A is moving ↓

$$2T \cdot 2 + T \cdot 4 - 2T \cdot V_A = 0$$

$$4 + 4 - 2V_A = 0$$

$$V_A = 4 m/s \downarrow$$

12. $l = I\omega = \frac{Pl}{2} = I\omega$

$$\frac{Pl}{2} = \frac{ml^2}{12}\omega$$

$$\omega = \frac{6P}{ml} \Rightarrow \quad \omega t = \frac{\pi}{2}$$

$$t = \frac{\pi}{2\omega} = \frac{\pi}{12P}(ml)$$

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13. Horizontal component of velocity remains constant.

$$u\sqrt{2}\times\frac{1}{\sqrt{2}}=v\sin 60^{o}$$

$$u=v\cdot\frac{\sqrt{3}}{2}\Rightarrow\left[v=\frac{2u}{\sqrt{3}}\right]$$

14. Change in momentum along y-direction will be cancelled out.

∴ Change in momentum along x-direction,

$$\Delta p = 2mv\cos\theta$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{\frac{\Delta p}{\Delta t}}{A}$$

$$= \frac{\Delta p}{A.\Delta t} = \frac{\Delta p}{A.\frac{L}{v\cos\theta}} = \frac{\Delta pv\cos\theta}{A.L}$$

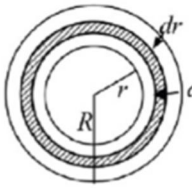
$$\text{Pressure } p' = \frac{2mv\cos\theta.\, v\cos\theta N}{V},$$

$$\because \left(n=\frac{N}{V}\right),\ (V = \text{Area}\times L = A\times L)$$

$$p' = 2mnv^2\cos^2\theta$$

15.

$$\text{Pressure } dp = \frac{dm.g}{A} = \frac{dm.g}{4\pi r^2} = \frac{\rho.4\pi r^2 drGM\frac{r}{R^3}}{4\pi r^2}$$



dm (*mass of elementary part*)

$$dp = \frac{\rho.4\pi r^2 drG.\ \rho.\frac{4\pi}{3}R^3\frac{r}{R^3}}{4\pi r^2} \Rightarrow dp = \frac{4\pi}{3}\rho^2 Grdr$$

$$\int dp = \int_r^R \frac{4\pi}{3}\rho^2 Grdr \Rightarrow p = \frac{4\pi}{3}\rho^2 G\left(\frac{r^2}{2}\right)_r^R$$

$$\Rightarrow p = \frac{4\pi}{3}\rho^2 G\left(\frac{R^2}{2}-\frac{r^2}{2}\right)$$

$$\Rightarrow p = \frac{4\pi}{3}\frac{\rho^2 G}{2}\left(R^2-r^2\right)$$

$$\Rightarrow p = \frac{2\pi}{3}\rho^2 G\left(R^2-r^2\right)$$

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16.

Magnetic flux through the loop $\phi = NBA$

induced e.m.f., $\varepsilon = -\dfrac{d\phi}{dt}$

and induced current

$$i = -\frac{1}{R}\frac{d\phi}{dt} = \frac{dQ}{dt} \Rightarrow -\frac{1}{R}d\phi = dQ$$

$$\text{Thus, } \frac{1}{20} \times (40 \times B \times 0.01) = 2 \times 10^{-5}$$

$$\Rightarrow B = 1 \times 10^{-3} T$$

17. Treat each ring as rod, emf induced in small ring and big ring is 4BVr each, emf s get added up as we move from top of one ring to other ring. Due to rotational motion no emf is generated.

18. Electrostatic potential energy

$$U \propto \frac{1}{r^3} \Rightarrow \frac{U_2}{U_1} = \frac{r_1^3}{r_2^3} = \frac{1}{8}$$

19. Zener diode is in parallel to load resistance and is connected in reverse bias.

20. AD is isochoric DC and BA are isothermal and CB is adiabatic.

21. $\mu = \tan\theta = \dfrac{dy}{dx} = \dfrac{x}{2} = \dfrac{1}{2}$

$x = 1$

$$y = \frac{x^2}{4} = \frac{1}{4} = 25\text{cm}$$

22. $$\frac{1}{f} = \left(\frac{\mu}{\mu_m} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\frac{1}{24} = (1.5 - 1)\left[\frac{2}{R}\right] \ldots\ldots\ldots (i)$$

$$\frac{1}{f'} = \left(\frac{1.5}{1.33} - 1\right)\left(\frac{2}{R}\right)$$

$$\frac{1}{f'} = \left(\frac{1.5 \times 3}{4} - 1\right)\frac{2}{R} \# (ii)$$

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(i) divided by (ii), $\frac{f'}{24} = 4$

$f' = 96$ cm

23. $P \propto V$

$\frac{P_1}{P_2} = \frac{V_1}{V_2} = \frac{V_o}{2V_o}$

$(P_2 = 2P_o)$

$\Delta Q = \Delta U + \Delta W$

$= \frac{f}{2} nR\Delta T + Area\ under\ P - v\ Graph$

$= \frac{3}{2}(4P_o v_o - P_o v_o) + \frac{1}{2} \times (P_o + 2P_o)(v_o)$

$= \frac{9}{2} P_o v_o + \frac{3}{2} P_o v_o = 6P_o v_o$

$\therefore \quad x = 6$

24. -dp/(dv/v)

25. $T = 2\pi\sqrt{\frac{l}{g}}$

$g = \frac{4\pi^2 l}{T^2}$

$\frac{\Delta g}{g} = \frac{\Delta l}{l} + 2\frac{\Delta T}{T}$

$\frac{\Delta g}{g} = (\frac{0.1}{10} + 2\frac{0.02}{0.5}) \times 100\%$

$\frac{\Delta g}{g} = 9\%$

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CHEMISTRY

26.

Molecular/ion	Number of unpaired electron	Bond order	Magnetic character
O_2^-	1	1.5	Para
O_2^+	1	2.5	Para
C_2^{-2}	0	3	Dia
N_2^{-4}	0	1	Dia

$$\text{Bond length} \propto \frac{1}{\text{Bond order}}$$

27. $r_n = r_0 \frac{n^2}{Z}$

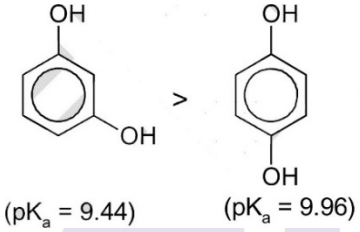
$$\Delta R = r_{n_2} - r_{n_1} = \frac{r_0}{Z}\left(n_2^2 - n_1^2\right)$$

$$He^+, \Delta R_1 = \frac{r_0}{2} \times \left(3^2 - 2^2\right) \ldots.(i)$$

$$Li^{+2}, \Delta R_2 = \frac{r_0}{3} \times \left(4^2 - 3^2\right) \ldots(ii)$$

$$\frac{\Delta R_1}{\Delta R_2} = \frac{3}{2} \times \frac{5}{7} = \frac{15}{14}$$

28.



(Correct acidic strength)

(pK_a = 9.44) (pK_a = 9.96)

29. $\beta - D - 2$ Deoxyribose sugar is present in DNA.

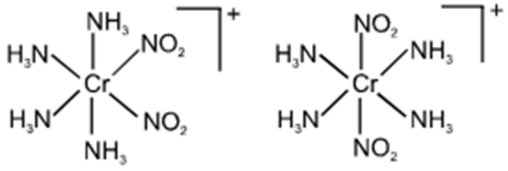
30. Ionisation isomerism

$[\overset{III}{Cr}(NH_3)_4(NO_2)_2](NO_3)$

and $[\overset{III}{Cr}(NH_3)_4(NO_2)(NO_3)](NO_2)$,

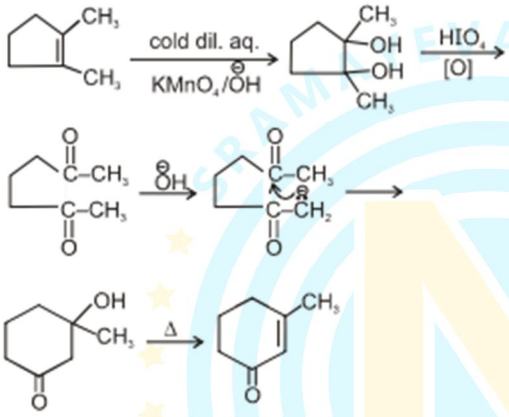
$[Cr(NH_3)_4(NO_2)_2]$ and $\left[Cr\left(NH_3\right)_4(ONO)_2\right]\left(NO_3\right)$ linkage isomerism

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cis– trans–

31.



32. $P(C_2H_4O)$ and $Q(C_3H_6O)$ must be carbonyl compounds as they both show +ve $NaHSO_3$ test. P must be ethanol as it gives haloform test.

$$P = CH_3 - \underset{\underset{O}{\|}}{C} - H, Q = CH_3 - CH_2 - CHO$$

$$CH_3 - CHO + CH_3 - CH_2 - CHO \xrightarrow[2.\ \Delta]{1.\ NaOH} CH_3 - CH = CH - CHO$$

$$+$$

$$CH_3 - CH_2 - \underset{\displaystyle CH_3}{\overset{|}{C}} = CH - CHO$$

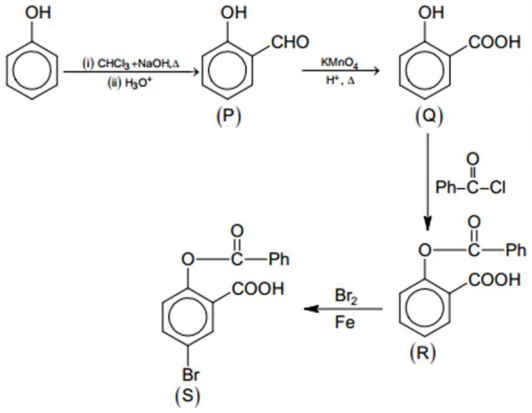
$$+$$

$$CH_3 - \underset{\displaystyle CH_3}{\overset{|}{C}} = CH - CHO$$

$$+$$

$$CH_3 - CH_2 - CH = CH - CHO$$

33.



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34.

$$CH_3CH_2OH \xrightarrow[H_2SO_4]{\text{Cold conc.}} \left[CH_3 - CH_2 - \overset{+}{O}H_2\right] HSO_4^-$$

Oxonium salt

$(i \simeq 2)$

$$Ph_3C - OH \xrightarrow[H_2SO_4]{\text{Cold conc.}} Ph_3C^+ + H_3O^+ + 2HSO_4^-$$

$(i \simeq 4)$

35. $\Delta T_f = k_f \times Molality$

$$\therefore \quad \frac{\Delta T_f}{k_f} = \text{molality} = \frac{\text{number of moles of glucose}}{\text{mass of water (in kg)}} = \frac{1}{1000} = \frac{\text{weight of glucose (in g)}}{180 \times 1}$$

$\therefore$ weight of glucose (in g) = 0.18 g

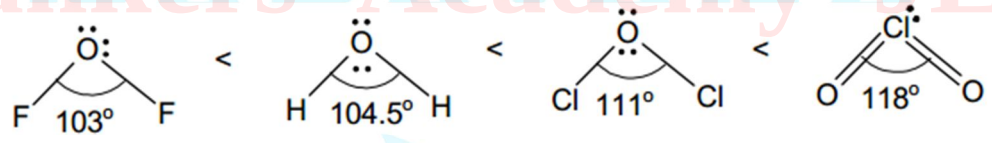
36. $$\frac{1}{x_o - a} - \frac{1}{x_o} = k_2 t$$

Rearranging,

$$\frac{a}{x_o - a} = x_o k_2 t$$

$\therefore \frac{a}{x_o - a}$ vs. t gives $x_o k_2$ as slope.

37.



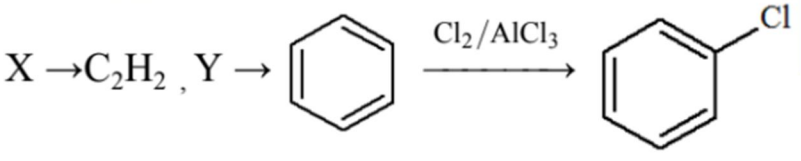
H_2O has larger bp – bp repulsion than OF_2 and OCl_2 has alrger steric repulsion between two large Cl atoms while in ClO_2 the bond angle is 118° (close to 120°.)

38. NCERT – XI, VOL – I, P.NO: 183

$\Delta = -10 \times 75.3 - 6030 - 36.8 \times 10$

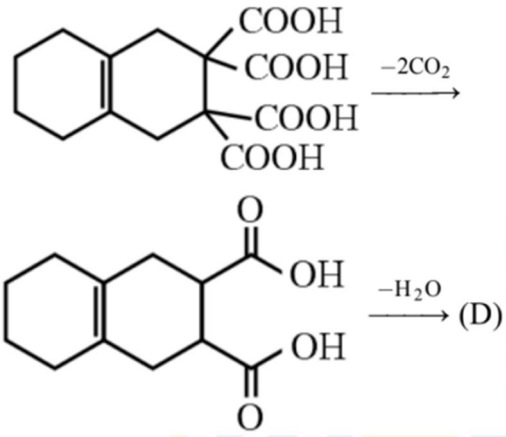
$= -753 - 6030 - 368 = -7151\,J$

39.



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40.



41. $\mu = \sqrt{n(n+2)}$

$1.73 = \sqrt{n(n+2)}$

On calculating the value of n we find n = 1

No. of calculating electrons = 1

Hence its configuration will be

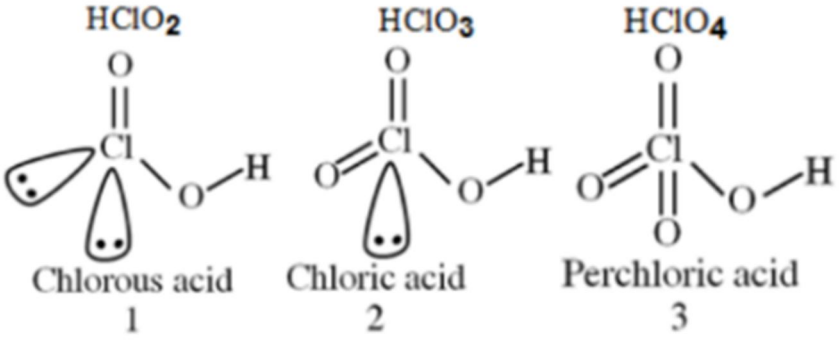
$V(23) = [Ar]3d^3 4s^2$

$V^{4+} = [Ar]3d^1$

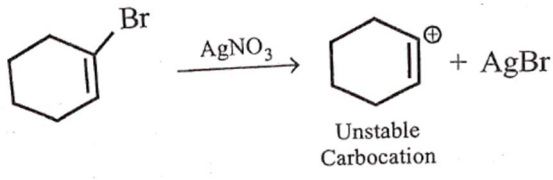
$\therefore$ Its chloride has the formula VCl_4

42.

Number of Cl = O bonds



43.



44. The reaction involved are

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$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$

$MgCO_3(s) \xrightarrow{\Delta} MgO(s) + CO_2(g)$

Let the weight of $CaCO_3$ be x g.

$\therefore$ Weight of $MgCO_3 = (2.21 - x)g$

Moles of $CaCO_3$ decomposed = Moles of CaO formed $= \frac{x}{100}$

$\therefore$ Weight of CaO formed $= \frac{x}{100} \times 56$

Moles of $MgCO_3$ decomposed = Moles of MgO formed $= \frac{2.21 - x}{84}$

$\therefore$ Weight of MgO formed $= \frac{2.21 - x}{84} \times 40$

$= \frac{2.21 - x}{84} \times 40 + \frac{x}{100} \times 56 = 1.152$

$\therefore \quad x = 1.1886g \approx 1.187g =$ Weight of $CaCO_3$

Weight of $MgCO_3 = 1.0214g \approx 1.023g$

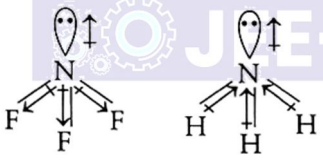
Hence, option (C) is correct.

45. Both Statement I and II are false. Their correct form are as follows

Statement I The dipole moment of NH_3 is higher than NF_3. The reason is that fluorine is more electronegative than nitrogen and thus pulls the electron towards itself. As number of NF bonds is 3, the three F-atoms pull electrons and hence resulting in decrease of dipole moment.

Statement II In NH_3 the orbital dipole due to lone pair and dipole moment of NH bond are in same

direction, but in NF_3 it will be in opposite direction.



Hence, option (D) is correct.

46. $AB_2(g) \longrightarrow A^+(g) + B^-(g) + B(g)$

$\Delta H_r = 2\Delta H_{A-B} + I.E._{(A)} + \Delta Heg(B)$

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$= 2\times100 + 250 - 150$

$= 200 + 250 - 150$

= 300 kcal/moles

So, x is 300 hence $\frac{x}{100}$ will be 3.

47. $q = \Delta U - W = -15 + \frac{1.2\times75\times100}{1000} = -6kJ$

48. Cathode : $Cu^{2+} + 2e^- \longrightarrow Cu$

Anode : $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

Moles of e^- used

$= \frac{Q}{F} = \frac{0.161\times5\times60}{96500} = 5\times10^{-4}$

Eq. of Cu^{2+} present

$= \frac{500\times0.1}{1000}\times2 = 0.1(\text{excess})$

$\therefore$ Moles of H+ produced = moles of

$e^- = 5\times10^{-4}$

$\therefore [H^+]_{final} = \frac{5\times10^{-4}}{500}\times1000 = 10^{-3}M \Rightarrow P^H = 3.0$

49.

$$H_2(4\text{ atm}) \longrightarrow 2H^+(C_1) + 2e^-$$
$$2H^+(C_2) + 2e^- \longrightarrow H_2(1\text{ atm})$$
$$H_2(4\text{ atm}) + 2H^+(C_2) \longrightarrow 2H^+(C_1) + H_2$$

For acetic acid,

$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{10^{-1}}} = 10^{-2}$

$[H^+](C_1) = C\alpha = 10^{-1}\times10^{-2} = 10^{-3}$

From NaOH $[H^+](C_2) = 10^{-11}$

$E = \frac{0.06}{2}\log\frac{C_2^2\times4}{C_1^2\times1}$

$= 0.03\log\left(\frac{10^{-22}\times4}{10^{-6}\times1}\right)$

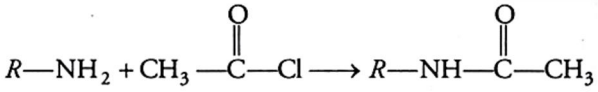
$= 0.03(\log4\times10^{-16})$

$= 0.03[-16 + 0.6]$

E = –0.462

50. The acetylation reaction is as follows

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Compound (X) molar mass =108 g/mol

Product molar mass =192 g mol^{-1}

Gain in molecular weight after acetylation with one $—NH_2$ group is 42.

Total increase in molecular weight = 84

$\therefore$Number of amino group in $X = \frac{84}{42} = 2$

Hence, 2 amino groups are present in compound X.

MATHEMATICS

51. Consider $\lim\limits_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)=L(say)$

If exist only when $\lim\limits_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)=0$ (i.e. it converts into 1^{∞} form), so the least degree of $f(x)$ would be 2 i.e., $f(x)=a_2x^2+a_3x^3+....$

$\because \quad \lim\limits_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)^{\frac{1}{x}}=e^2$

$\Rightarrow \quad e^{\lim\limits_{x\to 0}\left(\frac{x^2+f(x)}{x^2}\right)\times\frac{1}{x}}=e^2 \quad \Rightarrow \quad e^{\lim\limits_{x\to 0}\frac{x^2+f(x)}{x^3}}=e^2$

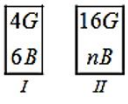
$\Rightarrow \quad \lim\limits_{x\to 0}\frac{x^2+f(x)}{x^3}=2 \quad \Rightarrow \quad \lim\limits_{x\to 0}\frac{x^2+a_2x^2+a_3x^3+...}{x^3}=2$

$\Rightarrow \quad a_2=-1, a_3=2$ and a_4, a_5 are any arbitrary constants. Since we want polynomial of least degree. Hence $f(x)=-x^2+2x^3$

$$=\lim_{x\to 0}\frac{-e^x(e^{x\cos x-x}-1)}{x\left(1+\frac{\sin x}{x}\right)}=\lim_{x\to 0}\frac{-e^x(e^{x\cos x-x}-1)(x\cos x-x)}{2x(x\cos x-x)}$$

$$=\lim_{x\to 0}\frac{(x\cos x-x)}{2x}=0$$

52.



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$$\frac{29}{50}=\frac{4}{10}.\frac{16}{n+16}+\frac{6}{10}.\frac{n}{n+16}\Rightarrow n=144 \text{ and } p=\left(\frac{3}{9}.\frac{15}{159}+\frac{6}{9}.\frac{144}{159}\right)=\frac{101}{159}$$

53. The appearance of $\frac{x}{y}$ in the given equation suggests the substitution $\frac{x}{y}=u$

$\Rightarrow dx=udy+ydu$

So, the given equation can be written as $\left(1+2e^{u}\right)\left(udy+ydu\right)+2e^{u}\left(1-u\right)dy=0$,

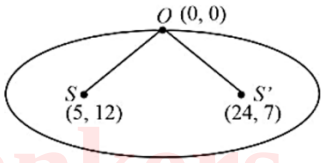
which is variable separable form.

$$\Rightarrow \quad \frac{\left(1+2e^{u}\right)}{u+2e^{u}}du+\frac{dy}{y}=0 \quad \Rightarrow \quad \int\left(\frac{1+2e^{u}}{u+2e^{u}}\right)du+\int\frac{dy}{y}=0$$

$$\Rightarrow \quad \log\left|u+2e^{u}\right|+\log\left|y\right|=\log\left|c\right| \quad \Rightarrow \quad \left(u+2e^{u}\right)y=c$$

$\Rightarrow \quad \left(x+2ye^{x/y}\right)=c$; comparing with given general solution, we get $\lambda=2$

54



$OS+OS'=38$

$SS'=38\times e$

$e=\frac{\sqrt{386}}{38}$

Hence eccentricity of hyperbola is $\sqrt{2}$. It is rectangular hyperbola $(a=b)$.

Now we know that, in case of hyperbola

$a^2e^2=a^2+b^2$ …(i)

$2ae=16$ (given) …(ii)

From (i) and (ii) equation of hyperbola is $x^2-y^2=32$

55. Variance $\left(\sigma^2\right)=\frac{1}{n}\sum_{i=1}^{n}\left(x_i-\bar{x}\right)^2$

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56.

(1,2,3) lie on

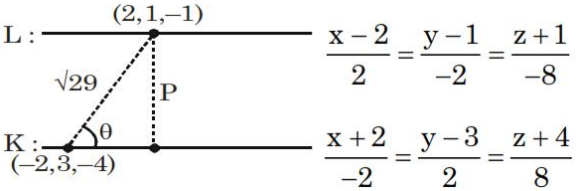
$$L \quad \therefore -\frac{1}{2}=\frac{1}{b}=\frac{4}{c}$$

$$\therefore b=-2,\ c=-8$$

Line L and k are parallel

$$\frac{2}{a}=\frac{b}{2}=\frac{c}{d}\Rightarrow\frac{2}{a}=-1=-\frac{8}{d}$$

$$\therefore a=-2, d=8$$



$$\frac{x-2}{2}=\frac{y-1}{-2}=\frac{z+1}{-8}$$

$$\frac{x+2}{-2}=\frac{y-3}{2}=\frac{z+4}{8}$$

$$\cos\theta=\left|\frac{8+4-24}{\sqrt{72}\sqrt{29}}\right|=\frac{\sqrt{2}}{\sqrt{29}}$$

$$\therefore \sin\theta=\frac{\sqrt{27}}{\sqrt{29}}$$

57.

Using LMVT on $f(x)$ for intervals [1,x] and [x,2]

$$\frac{f(x)-f(1)}{x-1}=f'(c_1)\Rightarrow\frac{f(x)-2}{x-1}\geq 1$$

$$\Rightarrow f(x)\geq x+1$$

and $$\frac{f(2)-f(x)}{2-x}=f'(c_2)\Rightarrow\frac{3-f(x)}{2-x}\geq 1$$

$\Rightarrow f(x)\leq x+1$

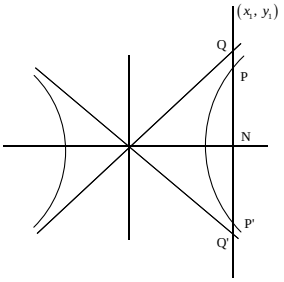
$\therefore f(x)=x+1$

$\therefore$ greatest valueof $g(x)=g(2)$

$$=\int_1^2(x+1)dx=\frac{5}{2}$$

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58.



$$Np = \frac{4}{5}\sqrt{x_1^2 - 25}$$

$$NQ = \frac{4}{5}x_1$$

$$pQ = \frac{4}{5}\left(x_1 - \sqrt{x_1^2 - 25}\right)$$

$$p'Q = \frac{4}{5}\left(x_1 + \sqrt{x_1^2 - 25}\right)$$

$$pQ.\,p'Q = 16$$

59. $y = mx - 2m - m^3 \Rightarrow m^3 + m(2-x) + y = 0$

$m_1 + m_2 + m_3 = 0$

$m_1 m_2 m_3 = -y$

$m_1 m_2 = -1$

$\Rightarrow m_3 = y$

$\Rightarrow m_1 + m_2 = -y$

$m_1 m_1 = 2 - x \Rightarrow -1 + y(m_1 + m_2) = 2 - x$

$\Rightarrow y^2 = (x-3)$

60. First arrange the letters of the word MONDAY in alphabetical order, i.e.,

ADMNOY

Total words beginning from A = 5!

Total words beginning from D = 5!

Total words beginning from MA = 4!

Total words beginning from MD = 4!

Total words beginning from MN = 4!

Total words beginning from MOA = 3!

Total words beginning from MOD = 3!

Total words beginning from MONA = 2!

Total words beginning from MONDAY = 1

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61. equation of tangent at θ is

$$\frac{x\sec\theta}{2}-\frac{y\tan\theta}{\sqrt{2}}=1$$

$$P=(2\cos\theta,0),Q=\left(0,-\sqrt{2}\cot\theta\right)$$

$$h=2\cos\theta \qquad k=-\sqrt{2}\cot\theta$$

Eliminating θ we get $\frac{4}{x^2}-\frac{2}{y^2}=1$

62.

$$\sum_{n=1}^{\infty}\sin^{-1}\frac{(2n+1)\left[\sqrt{n^2+2n}-\sqrt{n^2-1}\right]}{n(n+1)\left[\left(n^2+2n\right)-\left(n^2-1\right)\right]}$$

$$\sum_{n=1}^{\infty}\sin^{-1}\left[\frac{\sqrt{n^2+2n}-\sqrt{n^2-1}}{n(n+1)}\right]$$

$$\sum_{n=1}^{\infty}\sin^{-1}\left[\frac{\sqrt{(n+1)^2-1}-\sqrt{n^2-1}}{n(n+1)}\right]$$

$$\sum_{n=1}^{\infty}\sin^{-1}\left[\frac{1}{n}\sqrt{1-\frac{1}{(n+1)^2}}-\frac{1}{n+1}\sqrt{1-\frac{1}{n^2}}\right]$$

$$\sum_{n=1}^{\infty}\left(\sin^{-1}\frac{1}{n}-\sin^{-1}\frac{1}{n+1}\right)$$

$$\therefore S_n=\left(\sin^{-1}1-\sin^{-1}\frac{1}{2}\right)+\left(\sin^{-1}\frac{1}{2}-\sin^{-1}\frac{1}{3}\right)+$$

$$\left(\sin^{-1}\frac{1}{3}-\sin^{-1}\frac{1}{4}\right)+\cdots+\left(\sin^{-1}\frac{1}{n}-\sin^{-1}\frac{1}{n+1}\right)$$

$$Sn=\frac{\pi}{2}-\sin^{-1}\frac{1}{n+1} \qquad \lim_{n\to\infty}S_n=\frac{\pi}{2}.$$

63.

$$\frac{\alpha^2+\alpha+1}{\alpha^2-\alpha+1}=\frac{(\alpha-1)\left(\alpha^2+\alpha+1\right)}{(\alpha-1)\left(\alpha^2-\alpha+1\right)}$$

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$$=\frac{\alpha^3-1}{(\alpha-1)\left(\alpha^2-\alpha+1\right)}$$

$$=\frac{2\alpha^2-6\alpha}{\alpha^3-2\alpha^2+2\alpha-1}=\frac{2\alpha(\alpha-3)}{-4\alpha}=\frac{3-\alpha}{2}$$

Similarly remaining

$$\frac{\alpha(3-\alpha)+\beta(3-\beta)+\gamma(3-\gamma)}{2}$$

$$=\frac{3(\alpha+\beta+\gamma)-\left(\alpha^2+\beta^2+\gamma^2\right)}{2}=7$$

64.

$$I=\int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}\frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right)+\sin^{-1}\left(\frac{2x}{1+x^2}\right)}{e^x+1}=\frac{\pi}{2}\int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}\frac{1}{e^x+1}dx$$

$$\therefore 2I=\frac{\pi}{2}\cdot\int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}\left(\frac{1}{e^x+1}+\frac{e^x}{e^x+1}\right)dx\therefore I=\frac{\pi}{2\sqrt{3}}$$

65.

$$f(x)=\begin{cases}(x-3)(x+1) & e^{(3x-2)^2} & x\in(3,\infty)\\ -(x-3)(x+1) & e^{(3x-2)^2} & x\in[-1,3]\\ (x-3)(x+1) & e^{(3x-2)^2} & x\in(-\infty,-1)\end{cases}$$

Clearly non differentiable of $x=-1\&3$

66. $h\left(g\left(x\right)\right)=f\left(x\right)$

$\Rightarrow h\left(g\left(f\left(x\right)\right)\right)=f\left(f\left(x\right)\right)$

$\Rightarrow h(x)=f\left(f\left(x\right)\right)$ [using $g\left(f\left(x\right)\right)$]

$\Rightarrow h'(x)=f'\left(f\left(x\right)\right).f'(x)$

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Putting $x=1,$ we get

$h'(1)=f'(f(1)).f'(1)$

$=(3\times 36+3)\times 6=666$.

67. $I_2=\int_0^1\left(\frac{x}{5+x}\right)^{\frac{7}{2}}.\left(\frac{1-x}{5+x}\right)^{\frac{9}{2}}.\frac{dx}{(5+x)^2}$

Put $\frac{x}{5+x}=t\,\frac{5}{(5+x)^2}dx=dt\Rightarrow\frac{dx}{(5+x)^2}=\frac{1}{5}dt$

$\therefore I_2=\int_0^{\frac{1}{6}}(t)^{\frac{7}{2}}(1-6t)^{\frac{9}{2}}\frac{dt}{(5)^{11/12}}$; Now put $6t=\mu$ and simplify we get

$I_2=\frac{1}{5^{9/2}\times 6^{7/2}}I_1$ we Conclude a = 30

68. $\vec{c}=\lambda(\vec{a}\times\vec{b})$

or $\vec{c}.\vec{c}=\lambda(\vec{a}\times\vec{b}).\vec{c}$

$\Rightarrow$ $\frac{1}{3}=\lambda$

Also $|\vec{c}|^2=\lambda^2|\vec{a}\times\vec{b}|^2$

$\Rightarrow$ $\frac{1}{3}=\frac{1}{9}(a^2b^2\sin^2\theta)=\frac{1}{9}\times 2\times 3\sin^2\theta$

or $\sin^2\theta=\frac{1}{2}$

or $\theta=\frac{\pi}{4}$

69. $I=\int\frac{1}{\left[(x-1)^3(x+2)^5\right]^{1/4}}dx$

$\int\frac{1}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^2}dx$

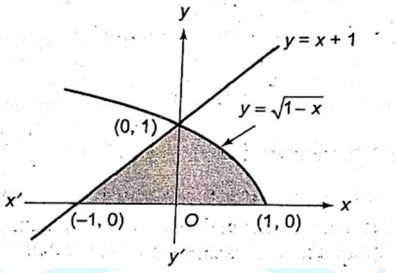
Let $\frac{x-1}{x+2}=t\Rightarrow\frac{3dx}{(x+2)^2}dt$

$\Rightarrow$ $I=\frac{1}{3}\int\frac{1}{t^{3/4}}dt,$

$=\frac{1}{3}\left(\frac{t^{1/4}}{1/4}\right)+C=\frac{4}{3}t^{1/4}+C=\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$

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70. Required area = shaded region



$= \int_0^1 (x_2 - x_1)\,dy$ (integrating along y-axis)

$= \int_0^1 \left[\left(1-y^2\right)-(y-1)\right]dy$

$= \frac{7}{6}\,sq.unit$

71. $\frac{\left(x^5+1\right)(1+x)^5}{x^5\left(x^4-x^3+x^2-x+1\right)} = \frac{(1+x)^6}{x^5}$

Constant term = coefficient of x^5 in $(1+x)^6 = {}^6C_5 = 6$

Coefficient of x^{-3} = coefficient of x^2 in $(1+x)^6 = {}^6C_5 = \frac{6\times5}{2} = 15$

72. The chord of contact from (h, k) to $y^2 = 4x$ is $ky = 2(x + h)$. If it touches the hyperbola

$\Rightarrow 4h^2 + k^2 = 4$ locus is $\frac{x^2}{1} + \frac{y^2}{4} = 1$

73.

$: |\alpha - 1 + i| = 2$ circle with $c(1,-1) r = 2$

$\therefore |\alpha|_{max} = \sqrt{2} + 2$

$|\beta - 4 + 5i| = 3$ circle with $c(4,-5)$ with $r = 3$

$(\beta)_{max} = \sqrt{41} + 3$

$\therefore 2|\beta|_{max} - 3|\alpha|_{max} = \sqrt{164} - \sqrt{18}$

$\therefore \left[\frac{k_1}{K_2}\right] = \left[\frac{164}{18}\right] = 9$

74. $|(\vec{a}\times\vec{b})\times\vec{c}| = |\vec{a}\times\vec{b}||\vec{c}|\sin 30°$

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$= \frac{1}{2}|\vec{a}\times\vec{b}||\vec{c}|$ (i)

We have, $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$

$\Rightarrow$ $\vec{a}\times\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$

or $|\vec{a}\times\vec{b}| = \sqrt{9} = 3$

Also given $|\vec{c} - \vec{a}| = 2\sqrt{2}$

Or $|\vec{c} - \vec{a}|^2 = 8$

Or $|\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a}.\vec{c} = 8$

Given $|\vec{a}| = 3$ and $\vec{a}.\vec{c} = |\vec{c}|$, using these we get

$|\vec{c}|^2 - 2|\vec{c}| + 1 = 0$

or $(|\vec{c}| - 1)^2 = 0$

or $|\vec{c}| = 1$

Substituting values of $|\vec{a}\times\vec{b}|$ and $\vec{c}$ in (i), we get

$|(\vec{a}\times\vec{b})\times\vec{c}| = \frac{1}{2}\times 3\times 1 = \frac{3}{2}$

So, $\frac{2}{3}\left|(\vec{a}\times\vec{b})\times\vec{c}\right| = 1$

75.

$$\frac{h-\lambda^2}{1} = \frac{k-2\lambda}{-1} = \frac{-2\left(\lambda^2 - 2\lambda + 1\right)}{2}$$

$h_1 = 2\lambda - 1; k = \lambda^2 + 1$

eliminating $\lambda, k = \left(\frac{h+1}{2}\right)^2 + 1 \quad 4K = h^2 + 2h + 3$

Locus is $(x+1)^2 = 4(y-1)$

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